

1.	Computer Networking A Top-Down Approach, James F. Kurose & Keith W. Ross, 8th edition, Pearson, 2022, ISBN-13: 978-0136681557
2.	Data and Computer Communications, 10th Edition, William Stallings, Pearson Education, 2013, ISBN: 0133506487, 9780133506488
3.	Larry L Peterson and Bruce S Davie, Computer Networks, fifth edition, ELSEVIER, 2011, ISBN-13: 978-0123850591.
E Books / Online Resources	
1.	https://www.tutorialspoint.com/data_communication_computer_network/index.html
2.	https://www.guru99.com/data-communication-computer-network-tutorial.html
MOOCs	
1.	https://nptel.ac.in/courses/106105081
2.	https://nptel.ac.in/courses/106105082
3.	https://onlinecourses.nptel.ac.in/noc22_ee61/
4.	https://www.digimat.in/nptel/courses/video/106105183/L01.html

OPERATING SYSTEMS				
Course Code:		CS2004-1	Course Type	PCC
Teaching Hours/Week (L: T: P)		3:0:2	Credits	04
Total Teaching Hours		40+0+26	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering				
Course Objectives:				
1.	Explain the concepts, principles and services of operating system.			
2.	Identify fundamental operating system concepts such as processes, inter-process communication, threads, CPU scheduling and demonstrate them.			
3.	Assess the need of concurrency and synchronization and apply them to write concurrent programs and analyze the cause for the occurrence of deadlocks and determine solutions to overcome the deadlocks.			
4.	Study the concepts of main memory and virtual memory allocation methods and demonstrate them.			
5.	Analyze the need for file system concepts, directory implementation and disk scheduling algorithms and demonstrate them.			
UNIT-I				
Operating System structure:				15 Hours
Operating System structure: Operating System Services, User and Operating System interface, System calls, System Services, Linkers and Loaders, Process Management: Process concept; Process scheduling; Operations on processes; Inter-process communication. Threads: Multithreading Models. CPU				

Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Thread scheduling																	
UNIT-II																	
Process Synchronization:													15 Hours				
Process Synchronization: The Critical section problem, Peterson's solution, Synchronization hardware, Semaphores, Monitors, Classical problems of synchronization. Deadlocks: System model; Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, avoidance, detection and recovery from deadlock. Main Memory: Paging, Structure of page table, Swapping.																	
UNIT-III																	
Virtual Memory:													10 Hours				
Virtual Memory: Demand paging, Copy-on-write, Page replacement, Allocation of frames. Implementing File System: File system Concepts, File System Structure, Operations and implementation, Directory implementation, Allocation methods, Free space management. disk scheduling algorithms																	
Suggested List of Experiments																	
1	Program to create five Child Processes using system call fork() and display their ids.																
2	Program to implement Scheduling algorithm																
3	Program to simulate producer-consumer problem.																
4	Program to demonstrate Page replacement algorithm to determine number of page faults.																
5	Program to demonstrate Bankers Deadlock avoidance algorithm.																
Course Outcomes: At the end of the course student will be able to																	
1.	Explain the structural components, services, system calls, creation and termination of processes operating systems.																
2.	Explain the multithreading concepts. Apply various CPU scheduling algorithms to optimize process execution time and CPU utilization.																
3.	Illustrate critical section problem, Peterson's solution, and semaphores and Apply the Banker's Algorithm to handle deadlock prevention by analyzing and ensuring safe resource allocation.																
4.	Demonstrate memory management techniques like paging, page table structure and swapping to optimize memory utilization.																
5.	Outline demand paging, copy on write and frame allocation techniques and Implement various page replacement algorithms. Apply various disk scheduling algorithms to optimize the storage performance.																
Course Outcomes Mapping with Program Outcomes & PSO																	
Program Outcomes→			1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes															1	2	3
CS2004-1.1			-	2	2	-	2	-	-	1	1	-	-	1	1	3	3
CS2004-1.2			-	2	2	-	2	-	-	1	1	-	-	1	1	3	3
CS2004-1.3			-	2	2	-	2	-	-	1	1	-	-	1	1	3	3
CS2004-1.4			-	2	2	-	2	-	-	1	1	-	-	1	1	3	3
CS2004-1.5			-	2	2	-	2	-	-	1	1	-	-	1	1	3	3
1: Low 2: Medium 3: High																	
TEXTBOOKS:																	
1	Operating System Concepts, Abraham Silberschatz, Greg Gagne, Peter B. Galvin, 12th Edition, John Wiley & Sons, 2018, ISBN: 9781119320913.																
REFERENCE BOOKS:																	
1.	D.M. Dhamdhare's "Operating Systems – A Concept-Based Approach" (2nd Edition, 2006, Tata McGraw-Hill)																

2.	P.C.P. Bhatt's "An Introduction to Operating Systems: Concepts and Practice" (5th Edition, 2019, PHI Learning)
3.	Harvey M Deital: Operating systems, 3rd Edition, Addison Wesley, 1990.
E Books / MOOCs/ NPTEL	
1.	http://www.uobabylon.edu.iq/download/M.S%2020132014/Operating_System_Concepts,_8th_Edition%5BA4%5D.pdf
2.	http://iips.icci.edu.iq/images/exam/Abraham-Silberschatz-Operating-System-Concepts--9th2012.12.pdf
3.	http://www.uobabylon.edu.iq/download/M.S%2020132014/Operating_System_Concepts,_8th_Edition%5BA4%5D.pdf
4.	https://freevideolectures.com/university/iit-bombay/
5.	https://www.cse.iitb.ac.in/~mythili/os/

MACHINE LEARNING			
Course Code:	CS2003-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+26	CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering			
Course Objectives:			
1.	Gain comprehensive knowledge about the role and significance of machine learning and various applications.		
2.	Create a model that can use to predict the class of the target variable by learning simple Regression and KNN algorithm.		
3.	Explore various supervised learning algorithms (like Decision tree, ANN, Naïve Bayes) and their usage in solving problems related to the current scenario.		
4.	Explore the various unsupervised learning algorithm to solve real world problems.		
5.	Evaluate the performance of the machine learning models using various metrics and Gain the knowledge about Reinforcement learning.		
UNIT-I			
			15 Hours
Introduction:			
Machine learning: what and why?- Types of machine learning, Supervised learning and Unsupervised learning.			
Applications of Machine Learning: Social Media Features, Product Recommendations, Image & Speech Recognition, Sentiment Analysis, Self-driving cars, Email Spam and Malware Filtering, Stock Market Trading, Medical Diagnosis, Online Fraud Detection, and Automatic language translation, Perspectives and Issues in Machine Learning.			
Supervised Learning-I:			